

Relative saliency affects attentional capture and suppression of color and face singleton distractors: evidence from event-related potential studies

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Prior research has yet to fully elucidate the impact of varying relative saliency between target and distractor on attentional capture and suppression, along with their underlying neural mechanisms, especially when social (e.g. face) and perceptual (e.g. color) information interchangeably serve as singleton targets or distractors, competing for attention in a search array. Here, we employed an additional singleton paradigm to investigate the effects of relative saliency on attentional capture (as assessed by N2pc) and suppression (as assessed by PD) of color or face singleton distractors in a visual search task by recording event-related potentials. We found that face singleton distractors with higher relative saliency induced stronger attentional processing. Furthermore, enhancing the physical saliency of colors using a bold color ring could enhance attentional processing toward color singleton distractors. Reducing the physical saliency of facial stimuli by blurring weakened attentional processing toward face singleton distractors; however, blurring enhanced attentional processing toward color singleton distractors because of the change in relative saliency. In conclusion, the attentional processes of singleton distractors are affected by their relative saliency to singleton targets, with higher relative saliency of singleton distractors resulting in stronger attentional capture and suppression; faces, however, exhibit some specificity in attentional capture and suppression due to high social saliency.

Key words: attentional capture; relative saliency; attentional suppression; N2pc; PD.

Introduction

Physically salient distractors appear to capture attention rapidly and automatically during visual search. For example, when searching for a friend in a crowd, an unfamiliar girl wearing a vibrant red dress may inadvertently enter the field of vision and capture the viewer's attention. Despite the attention-grabbing nature of the girl in the red dress, she is not what the individual was actively seeking. The viewer must suppress her as a distractor to efficiently locate her friend. However, the exact mechanism by which the saliency of a singleton distractor modulates attentional processes remains unknown (Melloni et al. 2012; Qu et al. 2017; Gaspelin and Luck 2019; Lamy 2021; Luck et al. 2021; Stilwell et al. 2023; Lin et al. 2024). Stimulus-driven and goal-driven theories hold fundamentally different assumptions regarding how physically salient stimuli influence attentional selection. The former theory posits that attentional selection is a passive process, driven by bottom-up physical salience, where salient stimuli automatically capture attention and are not influenced by cognitive control. In contrast, the latter argues that attentional selection is flexible and is under top-down cognitive control, meaning that attention can be directed toward task-relevant stimuli while simultaneously disregarding task-irrelevant stimuli (Folk et al. 1992).

Recently, signal suppression theory has appeared to offer a new perspective for resolving this dispute. This theory posits that salient stimuli can automatically capture attention. However,

bottom-up capture can be eliminated by employing top-down inhibitory mechanisms, providing a potential bridge between bottom-up and top-down theories (Sawaki and Luck 2010; Luck et al. 2021). Gaspelin et al. (2015) provided compelling evidence for this concept by introducing a capture-probe paradigm. In 70% of the trials, participants were tasked with searching for a specifically shaped target while ignoring a color singleton. In the remaining 30% of trials, letter probes were briefly presented inside the search elements, and participants were instructed to report as many letters as possible. Their findings indicated that when the participants engaged in a feature search strategy, the accuracy of the letter inside the irrelevant singleton distractor was reduced below that observed for letters inside the nonsingleton distractors. This observation serves as direct evidence supporting the idea that salient distractors can be ignored (Gaspar et al. 2016; Gaspelin et al. 2017; Stilwell et al. 2023).

However, Wang and Theeuwes (2020) raised a critique, suggesting that many prior studies examining signal suppression have used relatively small search array set sizes (e.g. four items), potentially resulting in color singletons with low saliency. According to their argument, if the saliency of the distractor is increased, the salient distractor may capture attention instead of being suppressed. To support this claim, these researchers used a procedure in which the task was similar to experiment 4 in Gaspelin et al. (2015), varying the search array set sizes (four, six, and ten items). They found no evidence for the suppression of singleton

distractor locations for an array size of six, but the effect was significantly reversed for a search array size of 10, indicating enhanced processing of probes presented at the singleton distractor location. This latter observation demonstrates that singleton distractors capture attention for larger search array sizes, even though participants use a feature search strategy to locate the target (Theeuwes 2023). This result provides evidence supporting the idea that stimuli with high salience can still capture attention, even when they are task irrelevant (Lin et al. 2024).

Whether top-down inhibitory mechanisms can prevent attentional capture when facing highly salient stimuli remains a vigorous debate. Studies to date examining this phenomenon have predominantly used single-feature stimuli and have been more automated in attentional processing. Numerous classic studies have indicated that color stimuli possess bottom-up physical salience and exhibit higher priority during attentional capture (Seiss et al. 2009); however, they can also be directly inhibited when serving as distractors (Gaspelin et al. 2017; Chang and Egeth 2021; Stilwell and Gaspelin 2021). However, it remains unclear whether stimuli with high social salience in daily life, such as faces or emotional information, can automatically capture attention and whether specificity in attentional suppression toward them exists. Faces, as significant social stimuli, encompass a wealth of social information (e.g. emotion, eye gaze, gender, and race) and are considered to possess higher saliency compared with other stimuli, making them more likely to capture attention (Kelly et al. 2019; Kauffmann et al. 2021). In visual search, faces appear to consistently “pop out,” and the pop-out effect for a face target is unaffected by the number of distractors in the background, which may be due to the automatic processing of faces (Golan et al. 2014; Prunty et al. 2023). Prior research has compared human faces with other stimuli, such as machines (Mayer et al. 2015), vehicles (Kauffmann et al. 2019), and animal faces (Simpson et al. 2019). These studies consistently found that human faces are processed more rapidly, showing a processing advantage in the face category. Eye-tracking evidence also confirms that faces are more easily detected and have a more remarkable ability to attract and sustain attention than other stimuli (Coutrot and Guyader 2014). Kauffmann et al. (2019) conducted a study in which participants were simultaneously presented with a human face and another image (e.g. a vehicle) on either side of the screen; results showed that the saccade response time was significantly faster when the target was a face (100 to 120 ms) than when other stimuli were presented. Furthermore, when faces and arrows were used as directional cues, faces exhibited stronger gaze direction effects than arrow cues. Moreover, when faces appeared as distractors, they interfered with the search for social and asocial nonface targets (Sato and Kawahara 2015; Ariga and Arihara 2018). In addition, unattended faces containing race, emotions, and configuration information can elicit a visual mismatch negativity component, which indicates the automatic processing of these features (Sun et al. 2022; Wang et al. 2022; Zeng et al. 2022). Finally, and most importantly, although still controversial, neuroimaging and electrophysiological results have often demonstrated specialized brain regions for processing face stimuli, such as the fusiform face area (Burns et al. 2019; Deen et al. 2020; Cohen et al. 2023) and event-related potential (ERP) component N170 (Bentin et al. 1996; Fu et al. 2012; Gao et al. 2019).

The aim of this study was to investigate whether a face, as an important social stimulus, would be similar in salience to other salient singletons (e.g. a color singleton) or would be as strong in salience as an abrupt onset due to the processing advantages mentioned above. Based on previous studies, it follows

that the salience of stimuli is a relative concept. The physical energy of a stimulus is determined by its physical attributes (e.g. brightness, orientation, thickness, and blurriness). For objects with similar physical energy, when placed in the search display, salience arises from the local contrasts between a stimulus and its surroundings (Koch and Ullman 1987; Duncan and Humphreys 1989; Nothdurft 1993). The contrast of salience between stimuli induces the notions of relatively high and low salience. Depending on the different reference points (i.e. singleton target, singleton distractor, or nonsingleton distractors) in the search array, the relative saliency between stimuli can be calculated in various ways. For example, the relative saliency of a singleton target can be calculated between itself and other nonsingleton distractors or between itself and a singleton distractor. In this additional singleton paradigm, we specifically emphasized relative saliency as the difference between the salience of a singleton target and that of a singleton distractor. Specifically, we manipulated the physical attributes of a singleton stimulus (either the thickness of a colored ring or the blurriness of a face) such that the attentional processes of the singleton distractor could be investigated as a function of its relative saliency to the singleton target.

The present study examined N2pc and P_D in ERPs as indicators of attentional capture and suppression, respectively, with color and face stimuli interchangeably set as the singleton target or distractor in the search display. Using an additional singleton paradigm, we investigated two main issues: first, whether relative saliency between a singleton distractor and a singleton target influences the attentional processes to the singleton distractor; and second, if so, whether there are differences in the attentional processes to a salient face and perceptual color as a function of their relative saliency when they mutually are the singleton target and distractor in a search array.

In Experiment 1, we predicted that a face singleton distractor in a color task may initially capture attention and then be subsequently suppressed, eliciting an N2pc, followed by a P_D. In contrast, a color singleton distractor in a face task may be directly suppressed or ignored, eliciting only P_D or no component. In Experiment 2, we enhanced the physical salience of color stimuli in the search array using bold color rings, thereby increasing the relative saliency of the color singleton. We expected that this manipulation would lead to enhanced attentional processes to the color singleton distractor, manifested by direct suppression (eliciting only a P_D) or the capture of attention first, with later suppression (eliciting an N2pc, followed by a P_D). In addition, we anticipated that the relative saliency of a face singleton would correspondingly decrease, leading to weakened attentional processes to a face singleton distractor (eliciting a smaller N2pc and a smaller P_D). In Experiment 3, we blurred face singletons in Experiment 2 to reduce their physical salience. We predicted that a face singleton distractor might be directly suppressed or ignored (eliciting only a P_D or no component). Conversely, a color singleton distractor with enhanced relative saliency would initially capture attention and then be subsequently suppressed (eliciting an N2pc, followed by a P_D).

Furthermore, on the basis of stimulus-driven theory (Theeuwes 1994; Wang and Theeuwes 2020; Lin et al. 2024), we speculated that the level of relative saliency would be associated with the degree of attentional capture and suppression. In particular, a singleton distractor with higher relative saliency was expected to elicit more robust attentional capture and suppression, thus resulting in larger amplitudes of the corresponding components. In contrast, we expected a singleton distractor with lower relative saliency to exhibit weaker or nonexistent attentional capture

and suppression, leading to smaller amplitudes or the absence of corresponding components. However, according to signal suppression theory (Gaspelin and Luck 2018; Stilwell et al. 2023), even a highly salient distractor can be proactively suppressed, with the level of relative saliency not necessarily influencing the attentional suppression of a singleton distractor.

Materials and methods

Participants

Participants were recruited from Guangzhou University and were compensated with a payment of approximately \$10 or a course credit. All participants were right-handed, had normal or corrected-to-normal vision, and reported no color blindness. Informed consent was obtained from all participants. The research protocol was approved by the local ethics committee of the School of Education, Guangzhou University. The sample size was estimated using G*Power 3.1 software, assuming a moderate effect size ($f = 0.25$), significance level of $\alpha = 0.05$, and a power of $1 - \beta = 0.80$. Each experiment included 24 participants (Experiment 1: 10 males, $M = 20.08$ years, standard deviation [SD] = 1.51; Experiment 2: 10 males, $M = 20.16$ years, $SD = 1.73$; Experiment 3: 11 males, $M = 19.80$ years, $SD = 1.20$). Participants with a 30% rejection rate due to blinks or eye movements were excluded, resulting in two exclusions in Experiment 1, four in Experiment 2, and four in Experiment 3.

Stimuli and procedure

Participants were seated in a dimly lit and electrically shielded room with a viewing distance of approximately 60 cm. Each experiment was implemented using E-Prime software (version 3.0, Psychology Software Tools, Inc., Pittsburgh, United States). A schematic illustration of the sequence of each experiment is shown in Fig. 1A. Every trial began with the presentation of a white fixation cross ($0.6^\circ \times 0.6^\circ$) at the center of the monitor for 500 ms. Participants were instructed to maintain their gaze on the fixation cross, avoiding head and eye movements until the end of the trial. Participants started each trial by pressing the spacebar using the thumb of the left or right hand. After the spacebar press, the search array was presented for 200 ms, and the maximum time for responding was 1800 ms. Upon providing a correct response, the participants proceeded to the subsequent trial without interruption. In the event of an incorrect response or failure to respond, a prompt message showing “Wrong” was displayed for 500 ms before moving on to the subsequent trial.

An example of the stimuli under the four conditions in the experiment is shown in Fig. 1B. The stimuli were displayed on a black background (Commission Internationale de l’Éclairage (CIE): 0.312/0.329, 1.0 cd/m^2) of a 17” cathode-ray tube computer monitor with a refresh rate of 60 Hz. The search arrays comprised six ellipses ($3.5^\circ \times 3.5^\circ$) tilted 45° to the left or right, arranged along a virtual circle 4.7° in diameter, and positioned in correspondence to the even-number locations on a clock face. The ellipses comprised neutral or scrambled faces overlaid with a color ring (experiment 1: $0.15^\circ \times 0.15^\circ$). The face materials (6 males and 6 females) were randomly selected from the Chinese Facial Affective Picture System (Gong et al. 2011), and 12 corresponding scrambled materials were generated. The color ring was randomly chosen from 12 colors (i.e. CIE: 0.39, 0.45; 0.31, 0.41; 0.25, 0.34; 0.53, 0.30; 0.50, 0.27; 0.37, 0.23; 0.48, 0.43; 0.52, 0.39; 0.54, 0.35; 0.21, 0.27; 0.23, 0.22; 0.28, 0.22; see Zhang and Luck 2008). The color and face luminance within the ellipses were balanced to maintain consistency ($20 \text{ cd}/\text{m}^2$). In addition, the color and face

singletons changed randomly across trials to prevent participants from attending to a specific face or color.

The display type could be divided into four conditions: lateral singleton target with no singleton distractor (LTND), midline singleton target with no singleton distractor (MTND), lateral singleton target with a midline singleton distractor (LTMD), and midline singleton target with a lateral singleton distractor (MTLD) according to the relative position of the singleton target and the singleton distractor. Participants were asked to report the orientation (leftward/rightward) of the target (face or color singleton) by pressing one of the two response keys (counterbalanced across participants) with the index or middle finger of their right hand (i.e. “Y” or “B”), with equal emphasis on response speed and accuracy. In the color task, participants were instructed to locate the ellipse with a unique color and determine its orientation, while the distractor was a face singleton. In the face task, participants were required to find the only ellipse with a face and determine its orientation, while the distractor was a color singleton. The remaining four nonsingleton distractors were scrambled faces with the same color as the singleton target. These two tasks were conducted in separate phases, with the order counterbalanced across participants. Stimuli in the search array were randomized in each trial.

Every experiment comprised 24 practice trials and 1,152 formal trials, organized into 24 blocks for color and face tasks (12 blocks for each task), with 48 trials for each block. Performance feedback, including mean accuracy and reaction times, was presented on the screen during the 30-s break after each block.

Each experiment employed the same equipment and procedure. The only modifications, as depicted in Fig. 1C, were as follows: in Experiment 2, a bold color ring ($0.58^\circ \times 0.58^\circ$) was overlaid on the ellipse stimuli; in Experiment 3, the face item was blurred using Gaussian blur (blur value: 6°) in Adobe Photoshop 2020, based on Experiment 2.

EEG recording and analysis

EEG activity was continuously recorded from 64 Ag/AgCl electrodes, positioned according to the 10–20 International System, using a Neuroscan Curry 8 system (Compumedics USA, Charlotte, NC, United States) set in AC mode, sampled at 1000 Hz, with 0.05 to 100 Hz bandpass filtering, using a Neuroscan Synamps digital amplifier system (Neuroscan Labs, Sterling, VA, United States). A vertical electrooculogram (VEOG) was recorded from two electrodes positioned 1.5 cm above and below the left eye. A horizontal electrooculogram (HEOG) was recorded using two electrodes on the outer canthi of both eyes. Using an electrode between FPz and Fz as the ground, EEG activity was referenced online to the left earlobe and then re-referenced offline to the average of the left and right earlobes. The electrode impedance was kept below 5 $\text{k}\Omega$ throughout the experiment.

Offline data analysis was performed using the EEGLAB toolbox (Delorme and Makeig 2004) and the ERPLAB plugin (Lopez-Calderon and Luck 2014) within the MATLAB environment. EEG data were bandpass filtered offline with a 0.1 Hz high-pass and 30 Hz low-pass, and the slope was set at 12 dB/octave using the pop_basicfilter function. Continuous EEG was segmented into epochs starting 200 ms before visual array onset and ending 800 ms after. EEG epochs were baseline corrected using the average activity in the time interval from -200 to 0 ms relative to the onset of the search array. After excluding trials associated with an incorrect response in the visual search task, individual trials were then excluded from the final analysis if they were contaminated by artifacts using the moving window peak-to-peak amplitude

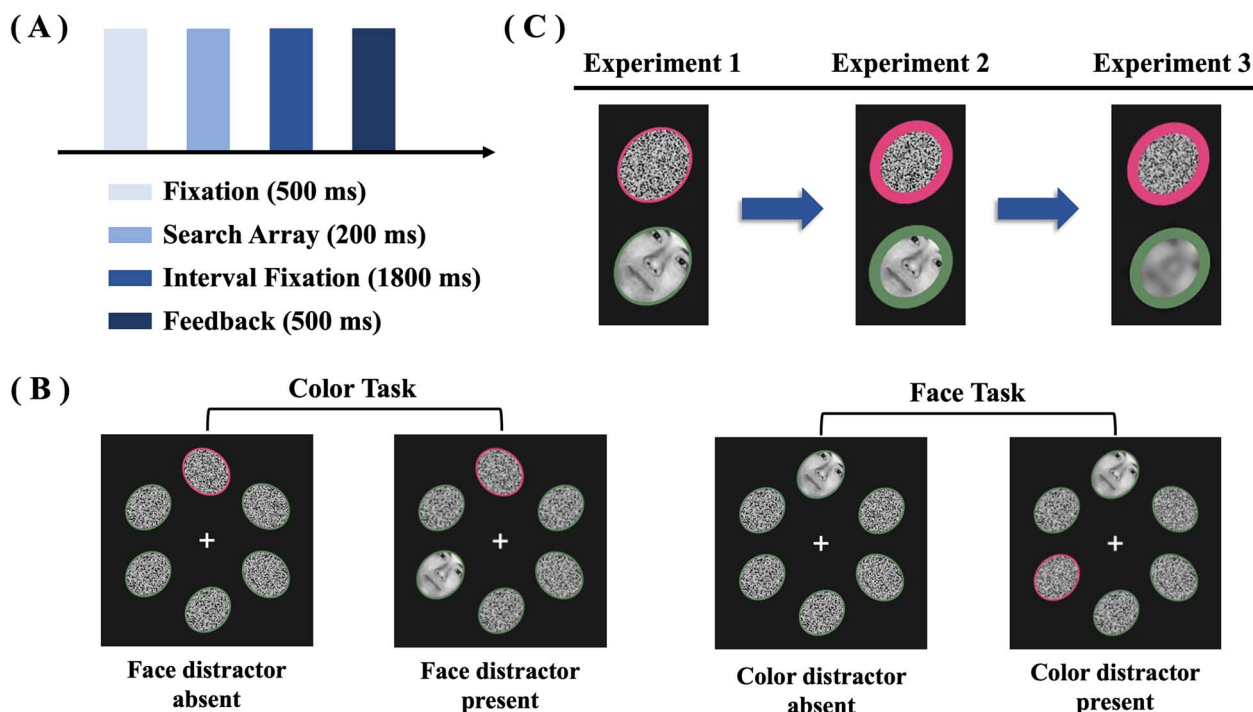


Fig. 1. Procedure, task, and stimuli for three experiments. A) A white fixation cross was presented for 500 ms, followed by a 200 ms search array. After the search array disappeared, participants had 1,800 ms to respond. Following a correct response, the next trial was initiated. In the case of an incorrect or no response, a prompt message was presented for 500 ms before proceeding to the next trial. B) The experiment consisted of two tasks. In the color task, participants were required to locate the unique ellipse with a specific color and determine its orientation. In the face task, participants were instructed to locate the unique ellipse with a face and determine its orientation. The order of the two tasks was counterbalanced across participants. Each task had two conditions, resulting in four conditions: color task with no face singleton distractor, color task with face singleton distractor, face task with no color singleton distractor, and face task with color singleton distractor. C) Experiment 1 consisted of ellipses ($3.5^\circ \times 3.5^\circ$) overlaid with color rings ($0.15^\circ \times 0.15^\circ$); Experiment 2 ($0.58^\circ \times 0.58^\circ$) bolded the color rings based on Experiment 1; and Experiment 3 was built upon Experiment 2 by introducing blurred face stimuli in the search array (blurring value: 6°).

method (VEOG deflection $\pm 50 \mu\text{V}$ within a time window of 150 ms; HEOG deflection $\pm 35 \mu\text{V}$ within a time window of 250 ms; or signal exceeding $\pm 80 \mu\text{V}$ within a time window of 200 ms anywhere in the epoch), resulting in an average rejection percentage of 9.17% (ranged between 0.18% and 26.27%) in Experiment 1, 9.66% (ranged between 0.45% and 27.82%) in Experiment 2, and 9.13% (ranged between 0.99% and 29.81%) in Experiment 3.

For laterally displayed singleton targets and distractors, the lateralized potential (i.e. N2pc, P_D) was computed as the average amplitude from contralateral-minus-ipsilateral difference waves relative to the side of the singleton targets and distractors. The presence of N2pc and P_D was assessed using one-tailed, one-sample t-tests against zero microvolts. According to prior studies, ERP amplitudes were computed from the PO7/8 electrode (e.g. Tay et al. 2019). ERPs elicited by the singleton target were measured in LTND and LTMD conditions, whereas ERPs elicited by the singleton distractor were measured only in the MTLD condition. The MTND condition was excluded from later statistical analyses because neither the singleton target nor the singleton distractor was in the lateral position, and no lateralized ERP component was expected. The time window for estimating the N2pc was 200 to 300 ms, which was chosen based on prior studies (Hickey et al. 2009; Luck and Gaspelin 2017). For the estimation of P_D , a “collapsed localizer” approach (Luck and Gaspelin 2017; Chen et al. 2023) was used to avoid biases in selecting the time period for analysis. The time window was defined as the onset and offset of P_D in the contralateral-minus-ipsilateral difference wave that was averaged across all trials (with latent voltages that reached 15% of the peak amplitude) based on fractional peak latency (Drisdelle

and Eimer 2021). According to this method, the time window used for P_D was 310 to 352 ms (Experiment 1), 262 to 333 ms (Experiment 2), and 249 to 338 ms (Experiment 3). In addition, the time window used for N2pc of the distractors was 193 to 244 ms in Experiment 3.

We conducted statistical analyses using R 4.1.2 (R Core Team 2023). The partial η^2 and Cohen’s d values are reported as effect sizes.

Results

Experiment 1

Behavioral results

Only correct responses were included in the analyses. Trials with RTs exceeding three SDs above/below the mean for each condition of a participant were considered outliers and excluded (7.79% of the total data). The average accuracy for the face task ($M = 0.97$, $SE = 0.004$) was higher than that for the color task ($M = 0.90$, $SE = 0.008$), $F(1, 23) = 76.12$, $P < 0.001$, $\eta_p^2 = 0.768$.

Manual Reaction Times (RTs) were used for a 2×2 repeated analysis of variance (ANOVA), considering task type (color vs. face task) and distractor condition (absent vs. present) as within-subject factors. As shown in Fig. 2A, the main effect of task type was significant. RTs were longer on the color task than on the face task (731 vs. 540 ms), $F(1, 23) = 136.99$, $P < 0.001$, $\eta_p^2 = 0.86$, indicating that the color task was more difficult. The main effect of the distractor condition was significant. RTs were longer when the singleton distractor was present than when it was absent (644

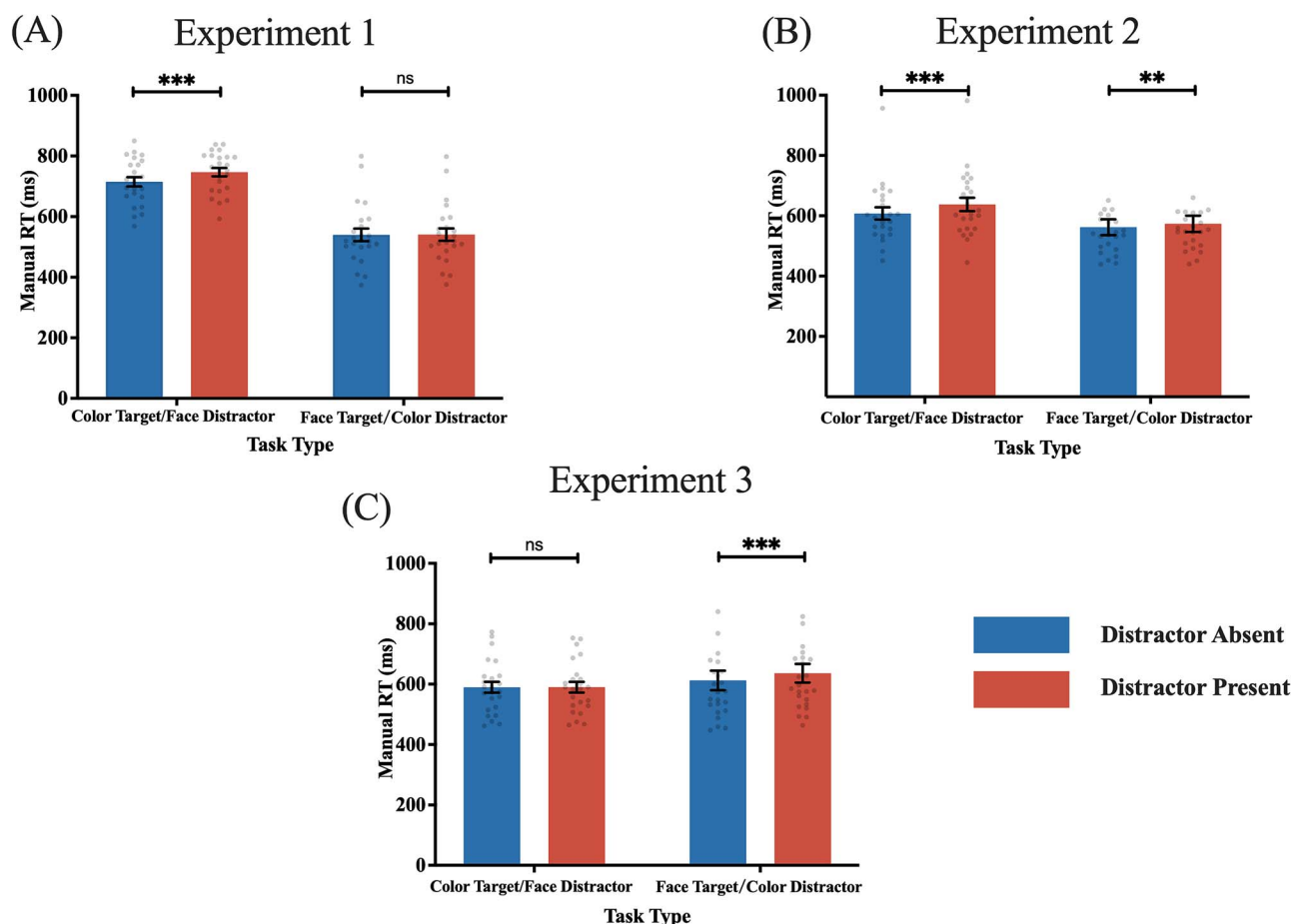


Fig. 2. Manual RT for experiments A) 1, B) 2, and C) 3 plotted as a function of task type (color vs. face task) and distractor condition (absent vs. present). For all panels, error bars represent the standard error of the mean RTs. The gray points depict individual participants.

vs. 627 ms), $F(1, 23) = 34.64$, $P < 0.001$, $\eta_p^2 = 0.60$, indicating the existence of a distractor interference effect. Notably, the interaction between task type and distractor condition was significant, $F(1, 23) = 27.01$, $P < 0.001$, $\eta_p^2 = 0.54$, with a more pronounced distractor interference effect for the face singleton distractor relative to the color singleton distractor (32 vs. 1 ms). Follow-up analysis showed that only the face singleton distractor exhibited an interference effect [747 vs. 715 ms, $t(23) = 6.06$, $P < 0.001$, Cohen's $d = 0.47$]. In contrast, the color distractor did not [541 vs. 540 ms, $t(23) = 0.48$, $P = 0.63$, Cohen's $d = 0.02$]. Moreover, in the absence of a distractor condition, RTs were longer on the color task than on the face task (715 vs. 540 ms), $t(23) = 10.73$, $P < 0.001$, Cohen's $d = 2.61$, indicating that the face singleton target was more salient than the color singleton target.

ERP results

N2pc for singleton targets

Average amplitudes within the 200 to 300 ms time window for LTND and LTMD conditions were used for the analysis of singleton target-elicited N2pc measures. The average amplitude of the contralateral N2pc wave was more negative than that of the ipsilateral wave. To formally analyze this, we computed the mean amplitude of N2pc waves elicited for face/color targets using contralateral-minus-ipsilateral difference waves. As shown in Fig. 3B (left panels), one-sample t -tests were conducted to assess whether each difference wave was significantly less than zero to confirm the presence of N2pc for face/color singleton

targets. N2pc was significantly elicited by both color singleton targets [$-0.30 \mu\text{V}$, $t(23) = -2.70$, $P = 0.013$, Cohen's $d = -0.55$] and face singleton targets [$-1.78 \mu\text{V}$, $t(23) = -8.97$, $P < 0.001$, Cohen's $d = -1.01$]. The amplitude of N2pc was significantly more negative for face singleton targets than for color singleton targets (-1.78 vs. $-0.30 \mu\text{V}$), $t(23) = -10.95$, $P < 0.001$, Cohen's $d = -2.23$. These results provide evidence that manipulation of physical salience was successful (i.e. the face singleton target was more easily found than the color singleton target). Scalp distribution maps for N2pc elicited by face and color singleton targets are shown in Fig. 3B (left panels).

N2pc and P_D for singleton distractors

Averaged amplitudes within the 200 to 300 ms time window for the MTLD condition were used to analyze singleton distractor-elicited N2pc measures. As shown in Fig. 4B (left panels), one-sample t -tests were conducted to assess whether each difference wave was significantly less than zero to confirm the presence of N2pc for face/color singleton distractors. The results revealed that N2pc was significantly elicited only by the face singleton distractor [$-0.55 \mu\text{V}$, $t(23) = -4.96$, $P < 0.001$, Cohen's $d = -1.01$], but not by the color singleton distractor [$0.03 \mu\text{V}$, $t(23) = 0.46$, $P = 0.650$, Cohen's $d = 0.09$]. Scalp distribution maps for N2pc elicited by face and color singleton distractors are shown in Fig. 4C (left panels).

It was anticipated that the singleton distractor stimulus, which was irrelevant to the task, would elicit a P_D component because we hypothesized that it would be suppressed. The average amplitude

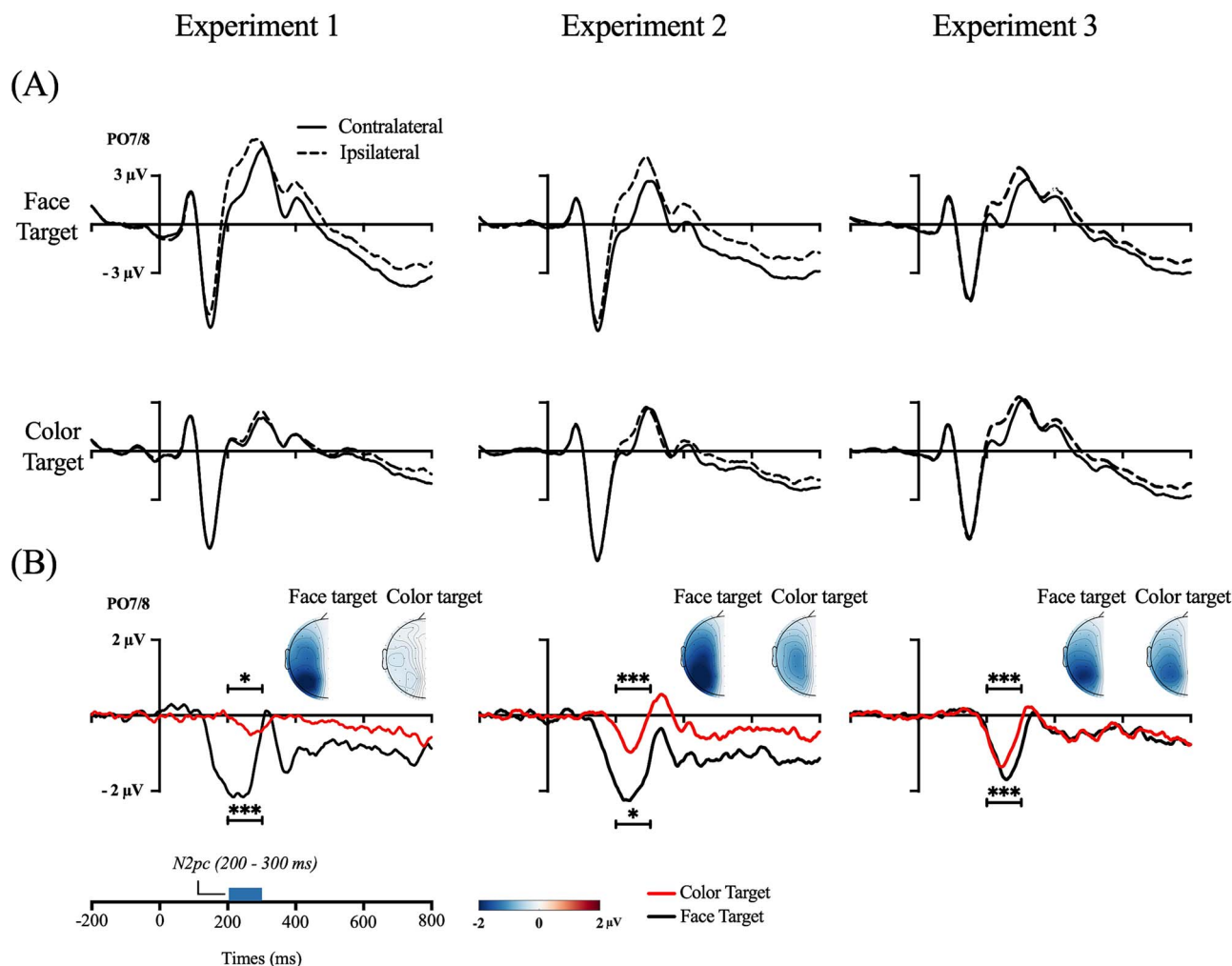


Fig. 3. Electrophysiological results from Experiments 1 (left panel), 2 (middle panel), and 3 (right panel) for singleton targets. A) Original waves of color and face singleton targets. The lines show grand-averaged contralateral (solid lines) and ipsilateral (dashed lines) ERPs elicited by lateral face singleton targets trials (upper panel) and color singleton trials (lower panel) at PO7/PO8. B) Difference (contra-ipsi) waves of color and face singleton targets and the scalp distribution maps for the N2pc elicited by color and face singleton targets. The grand mean topographical distribution of the difference wave is depicted in the upper right corner of the waveform plot.

of the P_D wave was more positive than that of the ipsilateral wave. The averaged amplitudes within the 310 to 352 ms time window for the MTLD condition were used for the analysis of singleton distractor-elicited P_D . To formally analyze this, we computed the mean amplitude of the P_D waves elicited for face/color distractors using contralateral-minus-ipsilateral difference waves. As shown in Fig. 4B (left panels), one-sample *t*-tests were conducted to assess whether each difference wave was significantly greater than zero to confirm the presence of P_D for face/color singleton distractors. The results revealed that P_D was significantly elicited only by face singleton distractors [$0.44 \mu V$, $t(23) = 3.31$, $P = 0.003$, Cohen's $d = 0.68$], but not by color singleton distractors [$-0.04 \mu V$, $t(23) = -0.58$, $P = 0.568$, Cohen's $d = 0.12$]. Scalp distribution maps for the P_D elicited by face and color singleton distractors are shown in Fig. 4C (left panels).

Experiment 2

Experiment 2 was designed to further enhance the physical saliency of color stimuli using bold color rings ($0.58^\circ \times 0.58^\circ$) compared with Experiment 1 ($0.15^\circ \times 0.15^\circ$), thereby increasing the relative saliency of color singleton distractors (relative to face singleton targets) within the search array. This allowed us

to examine whether the face singleton distractor's attentional processes change accordingly with the relative saliency change between face and color singletons in the search array. If the attentional processes toward face singleton distractors decreased with the increased saliency of color in Experiment 2, it would indicate a moderating effect of relative saliency. Compared with Experiment 1, we predicted that color singleton distractors with increased relative saliency in Experiment 2 would either be directly suppressed (i.e. eliciting only a P_D) or initially capture attention and subsequently be suppressed (i.e. eliciting an N2pc, followed by P_D). In contrast, a face singleton distractor with decreased relative saliency was expected to show reduced or no attentional capture or suppression effects (i.e. eliciting a smaller N2pc and a smaller P_D , or eliciting neither N2pc nor P_D).

Behavioral results

Only correct responses were included in the analyses. Trials with RTs exceeding three SDs above/below the mean for each condition of a participant were considered outliers and excluded (3.91% of the total data). The average accuracy for the face task ($M = 0.98$, $SE = 0.003$) was higher than that for the color task ($M = 0.97$, $SE = 0.004$), $F(1, 23) = 5.86$, $P = 0.024$, $\eta_p^2 = 0.20$.

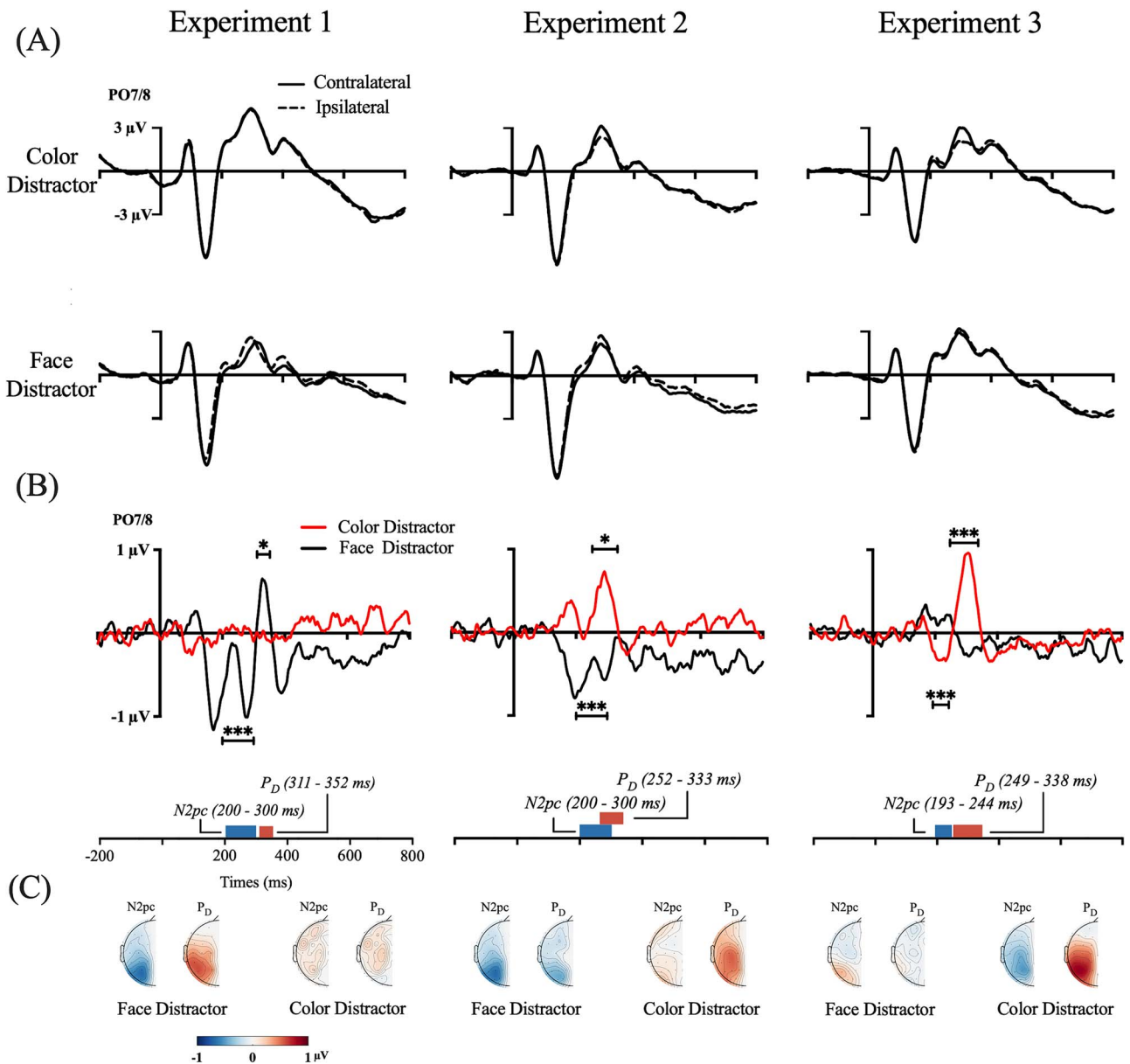


Fig. 4. Electrophysiological results from Experiments 1 (left panel), 2 (middle panel), and 3 (right panel) for singleton distractors. A) Original waves of color and face singleton distractors. The lines show grand-averaged contralateral (solid lines) and ipsilateral (dashed lines) ERPs elicited by lateral distractors for color singleton trials (upper panel) and face singleton trials (lower panel) at PO7/PO8. B) Difference (contra-ipsi) waves of color and face singleton distractors. C) Scalp distribution maps for the N2pc and PD elicited by face and color singleton distractors.

As shown in Fig. 2B, the main effect of task type was significant. RTs were longer on the color task than on the face task (623 vs. 568 ms), $F(1, 23) = 11.61$, $P = 0.002$, $\eta_p^2 = 0.34$, indicating that the color task was more difficult. The main effect of the distractor condition was significant. RTs were longer when the singleton distractor was present than when it was absent (606 vs. 585 ms), $F(1, 23) = 45.22$, $P < 0.001$, $\eta_p^2 = 0.66$, indicating a distractor interference effect. Notably, the interaction between task type and distractor condition was significant, $F(1, 23) = 7.44$, $P = 0.012$, $\eta_p^2 = 0.25$, with a more pronounced distractor interference effect for a face distractor relative to a color distractor (30 vs. 11 ms). Pairwise comparisons showed that both face and color distractors had interference effects [for the face singleton distractor, 638 vs. 608 ms, $t(23) = 5.48$, $P < 0.001$, Cohen's $d = 0.45$; for the color singleton distractor, 573 vs. 562 ms, $t(23) = 3.30$, $P = 0.003$, Cohen's $d = 0.17$]. Moreover, in the absence of the distractor condition,

RTs were longer for the color task than for the face task (608 vs. 562 ms), $t(23) = 2.96$, $P = 0.007$, Cohen's $d = 0.68$, indicating that the face singleton target was still more salient than the color singleton target.

Independent samples t-tests comparing Experiment 2 with Experiment 1 revealed that the interference effect produced by the color singleton distractor in Experiment 2 was significantly greater than that in Experiment 1 (11 vs. 1 ms), $t(46) = 2.47$, $P = 0.017$, and Cohen's $d = 0.71$. This indicates that increasing the relative saliency of the color enhanced the degree of interference effect to the face singleton target by the color singleton distractor. However, there was no significant difference in the interference effect of face singleton distractors between Experiments 2 and 1 (30 vs. 32 ms), $t(46) = -0.23$, $P = 0.823$, Cohen's $d = -0.07$, likely because the processing of the face singleton stimulus was automatic; even if the saliency of the color singleton target was

enhanced, the face singleton distractor still strongly interfered with the search for the color singleton target.

ERP results

N2pc for singleton targets

As shown in Fig. 3 (middle panels), one-sample *t*-tests were conducted to assess whether each difference wave was significantly less than zero to confirm the presence of N2pc for face/color singleton targets. N2pc was significantly elicited by both a color singleton target [$-0.60 \mu\text{V}$, $t(23) = -4.86$, $P < 0.001$, Cohen's $d = -0.99$] and a face singleton target [$-1.97 \mu\text{V}$, $t(23) = -6.75$, $P < 0.001$, Cohen's $d = -1.38$]. Crucially, the averaged amplitude of N2pc was significantly more negative for the face singleton target than for the color singleton target (-1.97 vs. $-0.60 \mu\text{V}$), $t(23) = -5.86$, $P < 0.001$, Cohen's $d = -1.20$. Scalp distribution maps for N2pc elicited by face and color singleton targets are shown in Fig. 3B (middle panels).

N2pc and P_D for singleton distractors

Averaged amplitudes within the 200–300 ms time window for the MTL condition were used to analyze singleton distractor-elicited N2pc measures. As shown in Fig. 4B (middle panels), one-sample *t*-tests were conducted to assess whether each difference wave was significantly less than zero to confirm the presence of N2pc for face/color singleton distractors. The results revealed that N2pc was significantly elicited only by face singleton distractors [$-0.51 \mu\text{V}$, $t(23) = -3.26$, $P = 0.003$, Cohen's $d = -0.67$], but not by color singleton distractors [$0.28 \mu\text{V}$, $t(23) = 3.39$, $P = 0.003$, Cohen's $d = 0.69$], although the latter had a component significantly different from zero, with a positively trending polarity. Scalp distribution maps for N2pc elicited by face and color singleton distractors are shown in Fig. 4 (middle panels).

An independent-sample *t*-test comparing the N2pc elicited by face singleton distractors in Experiments 1 and 2 revealed no significant difference (-0.55 vs. $-0.51 \mu\text{V}$), $t(46) = 0.18$, $P = 0.86$, Cohen's $d = 0.05$. No such comparison was conducted for color singleton distractors because no significant N2pc for color singleton distractors was observed in Experiment 2.

The averaged amplitudes within the 262 to 333 ms time window for the MTL condition were used for the analysis of singleton distractor-elicited P_D . As shown in Fig. 4 (middle panels), one-sample *t*-tests were conducted to assess whether each difference wave was significantly greater than zero to confirm the presence of P_D for face/color singleton distractors. The results revealed that P_D was significantly elicited only by color singleton distractors [$0.49 \mu\text{V}$, $t(23) = 4.00$, $P < 0.001$, Cohen's $d = 0.82$], but not by face singleton distractors [$-0.38 \mu\text{V}$, $t(23) = -2.64$, $P = 0.015$, Cohen's $d = -0.54$], although the latter had a component that significantly differed from zero, with a negatively trending polarity. Scalp distribution maps for the P_D elicited by face and color singleton distractors are shown in Fig. 4 (middle panels).

We further conducted an across-study analysis between Experiments 1 and 2 for singleton distractor-elicited P_D by applying independent-sample *t*-tests. P_D elicited by the color singleton distractor was more positive in Experiment 2 than in Experiment 1 (0.49 vs. $-0.04 \mu\text{V}$), $t(46) = 3.78$, $P < 0.001$, Cohen's $d = 1.09$, suggesting that increasing the relative saliency of color (i.e. using a bold color ring) led to an increase in the amplitude of P_D elicited by a color singleton distractor. However, P_D elicited by a face singleton distractor in Experiment 2 was less positive than that in Experiment 1 (-0.38 vs. $0.44 \mu\text{V}$), $t(46) = -4.19$, $P < 0.001$, Cohen's $d = -1.21$, which means that decreasing the relative saliency of face singleton distractors (because of increased color saliency)

resulted in a decrease in the amplitude of P_D elicited by face singleton distractors.

Experiment 3

In Experiment 3, face stimuli were blurred to reduce their saliency compared with Experiment 2 to further explore the main findings of Experiment 2. Consequently, compared with Experiment 1, the relative saliency of color stimuli in Experiment 3 was enhanced because of thicker color rings, whereas the relative saliency of face stimuli was decreased because of blurring of faces. Compared with Experiment 1, we predicted that color singleton distractors with enhanced relative saliency would capture attention first and then be suppressed (eliciting only a P_D or eliciting an N2pc followed by a P_D). In contrast, blurred face singleton distractors with reduced relative saliency would no longer possess specificity and thus exhibit weaker suppression without capturing attention (eliciting a smaller P_D or eliciting no component).

Behavioral results

Only trials with correct responses were included in the analyses. Trials with RTs exceeding three SDs above/below the mean for each condition of a participant were considered outliers and excluded (4.47% of the total data). The average accuracy for the face task ($M = 0.97$, $SE = 0.004$) and the color task ($M = 0.97$, $SE = 0.004$) was not significant [$F(1, 23) = 0.31$, $P = 0.581$, $\eta_p^2 = 0.01$].

Results in Fig. 2C show that the main effect of the distractor condition was significant. RTs were longer when the singleton distractor was present than when it was absent (613 vs. 601 ms), $F(1, 23) = 17.38$, $P < 0.001$, $\eta_p^2 = 0.43$, indicating that a distractor interference effect existed. Notably, the interaction between task type and distractor condition was significant, $F(1, 23) = 17.70$, $P < 0.001$, $\eta_p^2 = 0.44$, with a more pronounced interference effect for color singleton distractors relative to face singleton distractors (24 vs. 0 ms). Follow-up analysis showed that only color singleton distractors exerted an interference effect [(636 vs. 612 ms), $t(23) = 4.61$, $P < 0.001$, Cohen's $d = 0.25$]. In contrast, face distractors did not [(590 vs. 590 ms), $t(23) = 0.17$, $P = 0.87$, Cohen's $d = 0.004$]. However, the main effect of task type was not significant; RTs were not significantly different between the color and face tasks (590 vs. 624 ms), $F(1, 23) = 2.05$, $P = 0.166$, $\eta_p^2 = 0.08$.

We further compared distractor interference effects in Experiment 3 with those in Experiment 1, such that the role of relative saliency could be explored. The interference effects of color and face singleton distractors were subjected to a 2×2 repeated-measures ANOVA, considering task type (color vs. face task) as within-subject factors and group condition (Experiment 1 vs. Experiment 3) as between-subject factors. The interaction between task type and group condition was significant, $F(1, 46) = 44.43$, $P < 0.001$, $\eta_p^2 = 0.49$, indicating a pronounced change in the interference effect of color and face singleton distractors as a function of relative saliency. Pairwise comparisons showed that in Experiment 1, the interference effect of the face singleton distractor was greater than that of the color distractor (32 vs. 1 ms), $t(46) = 5.32$, $P < 0.001$, Cohen's $d = 1.10$, indicating that the face distractor with higher relative saliency caused a greater degree of attentional capture than the color distractor. In contrast, in Experiment 3, the interference effect of face singleton distractors was lower than that of color singleton distractors (0 vs. 24 ms), $t(46) = -4.10$, $P < 0.001$, Cohen's $d = -0.85$, suggesting that the decreased relative saliency of the face singleton distractor resulted in a lower degree of interference effect relative to the color singleton distractor with increased relative saliency. This demonstrates that a change in relative

saliency can result in a reversed interference effect for singleton distractors.

ERP results

N2pc for singleton targets

As shown in Fig. 3B (right panels), one-sample *t*-tests were conducted to assess whether each difference wave was significantly less than zero to confirm the presence of N2pc for face/color singleton targets. N2pc was significantly elicited by both color singleton targets [$-0.94 \mu\text{V}$, $t(23) = -6.65$, $P < 0.001$, Cohen's $d = -1.36$] and face singleton targets [$-1.24 \mu\text{V}$, $t(23) = -5.45$, $P < 0.001$, Cohen's $d = -1.11$]. Notably, the averaged amplitude of N2pc was not significantly different between the face and color singleton targets (-1.24 vs. $-0.94 \mu\text{V}$), $t(23) = -1.61$, $P = 0.120$, Cohen's $d = -0.33$. These results demonstrate that the physical saliency of color stimuli was successfully enhanced. Scalp distribution maps for N2pc elicited by face and color singleton targets are shown in Fig. 3B (right panels).

We conducted an across-study analysis between Experiments 3 and 1 for singleton target-elicited N2pc by applying independent-sample *t*-tests. The N2pc elicited by color singleton targets was more negative in Experiment 3 than in Experiment 1 (-0.94 vs. $-0.30 \mu\text{V}$), $t(46) = -3.54$, $P < 0.001$, Cohen's $d = -1.02$, demonstrating that the increased relative saliency of color singleton targets enhanced the degree of attentional capture. In contrast, no such difference was observed for face singleton targets between Experiments 3 and 1 (-1.24 vs. $-1.78 \mu\text{V}$), $t(46) = 1.79$, $P = 0.081$, Cohen's $d = 0.52$, indicating that face singleton targets can equivalently capture attention, even when blurred.

N2pc and P_D for singleton distractors

Averaged amplitudes within the 193 to 244 ms time window for the MTLD condition were used to analyze singleton distractor-elicited N2pc measures. As shown in Fig. 4B (right panels), one-sample *t*-tests were conducted to assess whether each difference wave was significantly less than zero to confirm the presence of N2pc for face/color singleton distractors. Results revealed that N2pc was significantly elicited only by color singleton distractors [$-0.30 \mu\text{V}$, $t(23) = -4.37$, $P < 0.001$, Cohen's $d = -0.89$], but not by face singleton distractors [$0.16 \mu\text{V}$, $t(23) = 1.60$, $P = 0.123$, Cohen's $d = 0.33$]. Scalp distribution maps for N2pc elicited by face and color singleton distractors are shown in Fig. 4C (right panels).

We further conducted an across-study analysis between Experiments 3 and 1 for singleton distractor-elicited N2pc by applying independent-sample *t*-tests. The N2pc elicited by face singleton distractors was less negative in Experiment 3 than in Experiment 1 (0.16 vs. $-0.55 \mu\text{V}$), $t(46) = 4.74$, $P < 0.001$, Cohen's $d = 1.37$, showing that the reduced relative saliency of color singleton targets decreased the degree of attentional capture. In contrast, color singleton distractors in Experiment 3 were more negative than those in Experiment 1 (-0.30 vs. $0.03 \mu\text{V}$), $t(46) = -3.40$, $P = 0.001$, Cohen's $d = -0.98$, indicating that when the relative saliency of color singleton distractors was increased, attentional capture was enhanced.

The averaged amplitudes within the 249 to 338 ms time window for the MTLD condition were used for the analysis of singleton distractor-elicited P_D measures. As shown in Fig. 4B (right panels), one-sample *t*-tests were conducted to assess whether each difference wave was significantly greater than zero to confirm the presence of P_D for face/color singleton distractors. The results revealed that P_D was significantly elicited only by color singleton distractors [$0.55 \mu\text{V}$, $t(23) = 4.05$, $P < 0.001$, Cohen's $d = 0.83$], but not by face singleton distractors [$-0.14 \mu\text{V}$, $t(23) = -1.34$, $P = 0.193$,

Cohen's $d = -0.27$]. Scalp distribution maps for the P_D elicited by face and color singleton distractors are shown in Fig. 4C (right panels).

We conducted an across-study analysis between Experiments 3 and 1 for singleton distractor-elicited P_D measures by applying independent-sample *t*-tests. P_D elicited by color singleton distractors in Experiment 3 was significantly more positive than that in Experiment 1 (0.55 vs. $-0.04 \mu\text{V}$), $t(46) = 3.88$, $P < 0.001$, Cohen's $d = 1.12$. However, P_D elicited by face singleton distractors in Experiment 3 was significantly less positive than that observed in Experiment 1 (-0.14 vs. $0.44 \mu\text{V}$), $t(46) = -3.45$, $P = 0.001$, Cohen's $d = -1.00$. This showed that the decreased relative saliency of a face singleton distractor reduced its interference with a color singleton target and required a lower degree of attentional suppression. Conversely, the increased relative saliency of a color singleton distractor enhanced its interference with a face singleton target, resulting in stronger attentional suppression.

Discussion

In this study, we employed an additional singleton task in a visual search paradigm to investigate the effects of relative saliency on attentional capture and suppression of color and face singleton distractors. Experiment 1 examined how the relative saliency between color and face singleton stimuli influences attentional processes. Experiment 2 enhanced the physical saliency of color stimuli compared with Experiment 1. Experiment 3 further reduced the physical saliency of face stimuli compared with Experiment 2 by blurring faces. The results consistently showed that the relative saliency between singleton targets and distractors influenced attentional capture and suppression processes. Specifically, higher relative saliency of a singleton distractor led to stronger attentional capture and suppression by that singleton distractor, as reflected by stronger N2pc and P_D amplitudes. Conversely, weaker or even absent N2pc and P_D components were observed when the relative saliency of a singleton distractor was lower. In addition, face stimuli showed some levels of specificity in attentional processes, likely because of their high social saliency.

Concept of relative saliency

Saliency of stimuli is a relative concept. For a single stimulus, its saliency usually refers to the local contrast between the stimulus and its background. Physical energy is related to saliency, but it is unequal to saliency per se. For example, the same white spot (having the same physical energy) could either have high saliency when presented against a black background or low saliency when presented against a gray background, indicating the relative nature of saliency. When placed in a search display, for objects with similar physical energy (e.g. brightness, orientation, thickness, blurriness), saliency arises from the local contrasts between a stimulus and its surroundings (Koch and Ullman 1987; Duncan and Humphreys 1989; Nothdurft 1993). The contrast of saliency between stimuli helps to explain the concepts of relatively high and low saliency.

In this study, local contrasts were manipulated by the thickness of color rings and the blurriness of faces so that the attentional processes of singleton distractors could be investigated as a function of their relative saliency to singleton targets. We specify the concept of "relative saliency" on the basis of a specific reference point and provide a clear operational definition for it. Under the present additional singleton paradigm, relative saliency was defined by the local contrasts between a singleton target and a

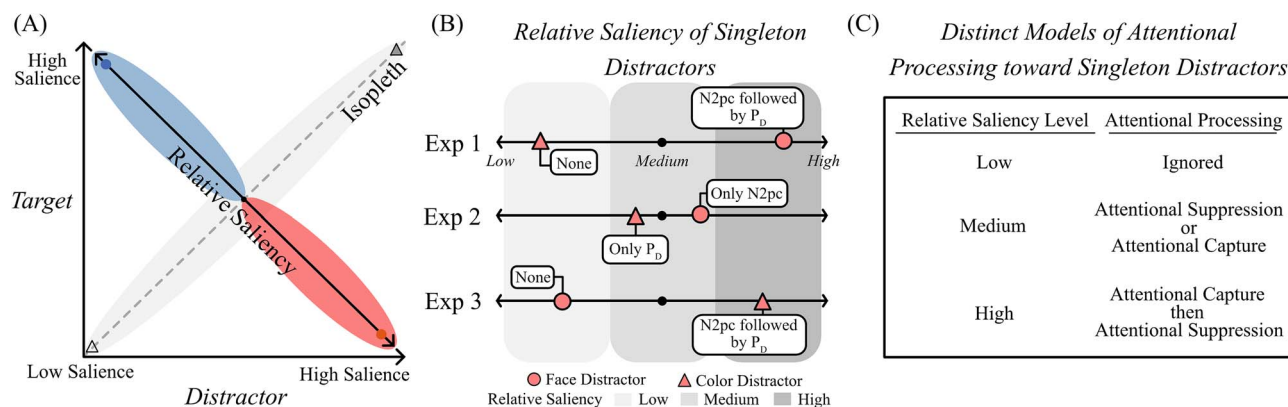


Fig. 5. Distinct models of attentional processing toward singleton distractor affected by relative saliency. A) Relative saliency changes with the saliency of the singleton target and distractor. The horizontal axis represents the saliency of the singleton distractor, with low saliency on the left and high saliency on the right. The vertical axis represents the saliency of the singleton target, with low saliency at the bottom and high saliency at the top. The diagonal axis at 135° represents the relative saliency of the singleton target and distractor, showing that the relative saliency for a singleton distractor is inversely proportional to the saliency of a singleton target, and vice versa. All points on the two-dimensional coordinates can be mapped onto the relative saliency line perpendicularly. The upper-shaded area in the diagonal axis indicates that the relative saliency of the singleton target is higher than the singleton distractor, with the dot within this area representing an extremely high relative saliency of the singleton target. Conversely, the lower-shaded area in the diagonal axis represents the relative saliency of the singleton distractor, which is higher than that of the singleton target, with the dot within this area representing the extremely high relative saliency of the singleton distractor. The relative saliency influences attentional processing toward the singleton distractor. However, it is crucial to note that the impact of relative saliency cannot be considered in isolation; task relevance, physical saliency, and even social saliency may also need to be considered. For instance, the 45° dotted line represents the isopleth line, on which the singleton target and the distractor have equal saliency. All points on the isopleth line are mapped to the central point on the relative saliency axis. Importantly, even if the relative saliency is the same, the attentional processing toward different singleton distractors may not be the same; in the position of the left-lower triangle, the singleton distractor has extremely low saliency, indicating that it is task-irrelevant and may be directly ignored. However, in the right-upper triangle's position, even if the singleton distractor is task-irrelevant, it may capture attention or be directly suppressed because of its extremely high saliency. In addition, even if the relative saliency line is symmetrical for the distractor and the target, as illustrated by the shaded areas in the diagonal axis, task relevance may bias attentional processing toward a singleton target due to top-down cognitive control (i.e. upper-shaded area is enhanced); a singleton distractor with high social saliency (e.g. a face distractor) may capture more attentional resource in a bottom-up manner (i.e. lower-shaded area is enhanced), because the social saliency may overlay with the physical saliency of the singleton distractor. B) Through different experimental manipulations, the changes in the relative saliency of color, face singleton distractors between three experiments, as well as the ERP components (i.e. N2pc or P_D) elicited by color and face singleton distractors with different relative saliency levels, were evaluated. The circle represents the face singleton distractor, and the triangle represents the color singleton distractor. The left background represents the relative saliency at a low level, the middle background at a medium level, and the right background at a high level. It is illustrated that the relative saliency of the face singleton distractor (the circles) is decreased from Experiment 1 (high) to Experiment 3 (low), whereas the relative saliency of the color singleton distractor (the triangles) is increased from Experiment 1 (low) to Experiment 3 (high). C) Different levels of relative saliency of a singleton distractor led to distinct models of attentional processing based upon (B). Specifically, when the relative saliency of a color or face singleton distractor is low, it is directly ignored during visual search and does not interfere with target detection, thus requiring no attentional suppression (i.e. it elicited no component). When the relative saliency is medium, it can only capture attention (i.e. it elicited only an N2pc) or be suppressed (i.e. it elicited only a P_D), with the P_D component occurring within or after the time window of the N2pc. When the relative saliency is high, it captures attention first and is subsequently suppressed (i.e. it elicited an enlarged N2pc followed by P_D).

singleton distractor within the search array (see Fig. 5A). Clearly, the relative saliency of a singleton distractor is negatively and linearly related to the saliency of a singleton target, and vice versa, as shown by the diagonal line in Fig. 5A. Moreover, it should be noted that relative saliency is not the only factor that affects the attentional process, as shown by the gray and colored areas in Fig. 5A. For example, the saliency of a singleton target and a singleton distractor may affect attentional capture and/or suppression, even though the relative saliency of a singleton distractor is low. Task relevance may also play a critical role in the top-down cognitive control factor, which biases attentional processing on the target relative to the distractor end, as illustrated by colored areas on the relative saliency coordinate in Fig. 5A. Additionally, high social saliency (e.g. a singleton face) may play a specific role and contribute to relative saliency, which is typically based on physical saliency of singleton targets and distractors; however, this point has yet to be fully elucidated.

Several previous studies have manipulated the saliency of stimuli in the additional singleton paradigm using different reference points for saliency computations. For example, Barras and Kerzel (2017a) manipulated saliency by contrasting the similarity between targets and nontargets (Experiment 1) and the

similarity between distractors and nontargets (Experiment 2). In their study, when the shape of a target/distractor was square, the saliency (shape-based similarity) was high when nontargets were diamonds and low when they were circles (see also Barras and Kerzel 2017b). Chen et al. (2023) used stimuli with different saliency levels (e.g. color, onset singleton). They manipulated the saliency based on the contrast between different singleton distractors (e.g. color vs. onset singleton distractors). Gaspar et al. (2016) manipulated saliency on the basis of the local contrast between nontargets and each color singleton distractor, and this local contrast was calculated as the distance in CIE. Saliency was high in situations where the color value of singleton distractors was far away from that of nontarget items, and it was low in situations where the color value of singleton distractors was close to that of nontarget items in CIE color space.

It is worth noting that, unlike the studies mentioned above that manipulated saliency with stimuli of the same dimension (e.g. shape or color dimension), we used ellipses consisting of neutral or scrambled faces overlaid with a color ring as stimuli, which involved a high social saliency stimulus category (face) and cross-dimensional (i.e. face vs. color) manipulations. Furthermore, we introduced color and face stimuli with varying levels

of physical properties into the same search array, where they served as either a singleton target or a distractor, competing for attentional resources. Therefore, the reference point used in calculating saliency in the current study was different because it was calculated by the local contrast between a singleton target and a singleton distractor. Second, in the previous studies mentioned above, the physical properties of the target and distractors were fixed. However, the current study manipulated the physical properties of the stimulus by changing the thickness of the color ring (Experiment 2) and the blurring of faces (Experiment 3), enhancing the local contrast between the singleton target and the singleton distractor to influence the attentional processes of a singleton distractor, as a function of relative saliency.

The time course of N2pc and P_D components may be affected by relative saliency

Based on findings from prior studies, the P_D component has been proposed to measure the attentional suppression of specific distractors and has served as a crucial ERP index for comprehending the attentional processing of salient distractors. The P_D component can manifest in the time range of 100 to 500 ms, reflecting the fact that suppression can occur either before or after the initial shift of attention (Hickey et al. 2009; Gaspelin et al. 2023). An early occurrence of P_D (e.g. 100 to 275 ms), before or during the N2pc time window, would be interpreted as a suppressive process that proactively prevents attentional allocation to distractor stimuli (Wang et al. 2019). Conversely, a late occurrence of P_D (e.g. 275 to 500 ms) would be interpreted as a suppressive process involved in terminating attentional allocation to a stimulus (Liesefeld et al. 2017; Drisdelle and Eimer 2023).

Our systematic manipulation of the physical properties of stimuli revealed that the time sequence of N2pc and P_D may be altered as a function of relative saliency and stimulus type. Specifically, color singleton distractors in Experiment 1 had lower relative saliency than face singleton targets and were directly ignored during visual search, without interfering with the search for the face singleton targets. This resulted in the absence of any later-ized ERP components (i.e. neither N2pc nor P_D) associated with attentional capture or suppression (see also Gaspar et al. 2016). In Experiment 2, when the saliency of color was enhanced, the color singleton distractor with increased relative saliency could no longer be ignored, even when it was task irrelevant. Although a 10-ms interference effect was observed in behavioral data, only the P_D component (262 to 333 ms) was observed in ERP. According to the fleeting capture hypothesis (Theeuwes 2010), attention can be directed to the distractor location and subsequently redirected to the target location without eliciting a distractor N2pc. However, Jannati et al. (2013) evaluated and opposed this “fleeting capture hypothesis,” proposing that the small interference effect may be linked to spreading inhibition (Gaspar and McDonald 2014) and nonspatial filtering (Folk and Remington 1998). For example, when a singleton distractor is present, there is a 10 ms cost compared to when it is absent. It could be that the distractor captured attention or that the participant’s attention initially focused on the target stimulus, but the presence of the distractor slowed down the speed at which attention shifted to the target. We cannot definitively conclude that the observed 10 ms cost is necessarily due to attentional capture by a distractor. Thus, this relatively late P_D component without N2pc may represent suppression without prior attentional capture. In Experiment 3, based on the findings of Experiment 2, the physical properties of the faces were reduced, further enhancing the relative saliency of the color singleton distractors. This resulted in N2pc being elicited first, followed

by the P_D component, indicating that color singleton distractors initially captured attention and were subsequently suppressed (Theeuwes 1994; Rodríguez Holguín et al. 2009; Ruthruff et al. 2020).

Face singleton distractors had a higher relative saliency in Experiment 1. The results showed that face singleton distractors elicited an N2pc followed by a P_D, indicating that they initially captured attention and were subsequently suppressed. In Experiment 2, although the relative saliency of face singleton distractors was reduced, they still induced a significant attentional capture effect. This may be attributed to the specificity of the automatic processing of faces (Golan et al. 2014; Prunty et al. 2023), as the relative saliency did not seem to affect the attentional capture process of face singleton distractors. However, it is worth noting that the P_D component was not elicited in this case, indicating that after reducing the relative saliency of face singleton distractors, they could be processed without the need to allocate cognitive resources for suppression. In other words, when the relative saliency of a singleton target reaches a certain level, participants can directly focus on processing the target without the need to suppress distractors to complete the search task (Gaspar and McDonald 2014; Zhao et al. 2023). This indicates that changes in relative saliency have differential effects on the attentional capture and suppression processing of a face singleton distractor relative to a color singleton distractor. Furthermore, in Experiment 3, face stimuli were blurred to reduce their relative saliency during visual search. Results showed that blurred face singleton distractors no longer interfered behaviorally with the search for color singleton targets and did not elicit any N2pc or P_D components, indicating that face singleton distractors with lower relative saliency could be ignored, possibly because they no longer served as important pieces of social information requiring rapid recognition or automatic processing (Fan et al. 2014; McCourt et al. 2023).

It should also be noted that there have been some criticisms of P_D. In brief, it has been proposed that the P_D is nothing more than a contralateral N2pc to a nontarget (Kerzel and Burra 2020; Kerzel and Huynh 2023). For example, Kerzel and Huynh (2023) presented all possible placements of the target and distractor (adjacent + opposite) in Experiment 1, whereas they presented only the adjacent placement in Experiment 2. Moreover, they set up a cue color (i.e. the distractor) at the beginning of the trial, expecting participants to suppress the stimulus in the distractor cue color to locate the target. They found that positive responses to the distractor were followed by positivity in the context of “adjacent + opposite” placements in Experiment 1. They ascertained that the observed late contralateral positivity was N2pc, instead of P_D, indexing attentional deployment to the nontarget on one side. Their rationale was that participants readily adopted a strategy to attend a nontarget opposite the distractor, which can avoid the distractor. Moreover, participants exhibited a horizontal advantage for scanning relative to vertical stimuli, as they responded faster to targets on the horizontal display (Kerzel and Burra 2020).

Their study seems to support the notion that top-down cognitive control (such as strategy or pre-cueing) can modulate the bottom-up attentional capture and suppression elicited by the low-level properties of a salient distractor. However, the results of this study seem unlikely to support this hypothesis. In our research, it was difficult for participants to adopt a strategy of attending a nontarget opposite the distractor to avoid the distractor, as they were not informed of the color of the distractor via pre-cueing. Furthermore, the difference between the N2pc and P_D components, as a function of relative saliency, was observed under

the same task instruction. Moreover, we did not observe a lateral advantage for scanning horizontal targets over vertical targets in the three experiments (see also [Chen et al. 2023](#)). It is likely that the differences in experimental design, stimuli, and task between these studies may account for the discrepancy observed across studies in the time course of N2pc and P_D .

Different levels of relative saliency of singleton distractors leads to distinct models of attentional processing

Consistent with the hypothesis of relative saliency, the results of this study demonstrated that when the singleton target was a face with higher relative saliency, a color singleton distractor with lower relative saliency exerted minimal or even no interference in the search for the face singleton target. However, when the singleton target was a color with lower relative saliency, a face singleton distractor with higher relative saliency significantly interfered with the recognition of the color singleton target. Interestingly, subsequent experiments (Experiments 2 and 3) revealed that a color singleton distractor with enhanced relative saliency exerted a greater interference effect on the search for a face singleton target. In contrast, a face singleton distractor with decreased relative saliency reduced or even eliminated the interference effect on the recognition of a color singleton target. On the basis of the aforementioned results, we propose that when employed with an additional singleton paradigm, if the singleton distractor is more salient than the singleton target, the attentional processing of the singleton distractor involves an initial stage of attentional capture (N2pc), followed by a stage of attentional suppression (P_D).

First, each stage is characterized by distinct components that represent their respective processes ([Chen et al. 2023](#)); however, the time courses of these components are likely variable. Second, given the singleton detection mode of visual search, the order of these stages cannot be altered, which means that the time window of P_D is later than the N2pc time window ([Chen et al. 2023](#)). In this mode, color and face singletons were changed randomly across trials to prevent participants from attending to a specific face or color. Participants needed to use a parallel search strategy to extend their attention to encompass all items in the search display. In such conditions, the initial search simultaneously unfolds across all items to locate the most salient item in the search display ([Theeuwes 2023](#)). However, several previous studies have shown that when employing other visual search modes (e.g. feature search mode), a singleton distractor can be directly suppressed (i.e. only eliciting a P_D). For example, [Gaspelin and Luck \(2018\)](#) conducted an experiment in which participants were tasked with a target shape (e.g. green diamond) in a display of heterogeneous stimuli (Experiment 1), which encouraged participants to employ a feature search mode. Their findings revealed that a lateral singleton distractor only elicited the P_D , indicating it could be directly suppressed without capturing attention. This might be attributed to the fact that, in their studies, even though the target was a singleton stimulus, it was fixed (e.g. find a green diamond). This might lead participants to concentrate on the fixed feature of the target by employing the feature search mode rather than bottom-up saliency. In this mode, the search transforms into a sequential search, consequently diminishing the propensity for salient distractors to capture attention (see also [McDonald et al. 2013](#); [Gaspar and McDonald 2014](#); [Stilwell et al. 2022](#)). It is important to note that they utilized heterogeneous search arrays with only four elements; the concern arises that, even with a display containing only a few items, uniqueness does not necessarily equate to the saliency of the distractor. This

raises questions about the suppression of a salient distractor when none of the elements are salient enough to conclude that the singleton distractor could be directly suppressed ([Wang and Theeuwes 2020](#)). Third, if the relative saliency between a target and a distractor is approximately the same, one of the stages may not exist (i.e. there is only N2pc or only P_D). In addition, if the singleton distractor is less salient than the target, it would be directly ignored without eliciting any component (i.e. neither N2pc nor P_D ; see also [Gaspar et al. 2016](#)).

Depending on the relative saliency of the singleton distractor, three processing modes may exist for its attentional processing (see [Fig. 5C](#)). When the relative saliency of the singleton distractor is at a low level (i.e. significantly lower than the singleton target), it can be directly ignored without interfering with the search for the singleton target, thus requiring no attentional suppression (i.e. eliciting neither N2pc nor P_D component). When the relative saliency of the singleton distractor is at a medium level (i.e. the relative saliency with the singleton target is about the same), it can only capture attention (elicited only N2pc) or be suppressed (elicited only a P_D), with the P_D component occurring within or after the time window of the N2pc. When the relative saliency of a singleton distractor is at a high level (i.e. significantly higher than the target), it captures attention first and must subsequently be suppressed (eliciting an enlarged N2pc and P_D). In other words, the relative saliency between a singleton distractor and a singleton target determines the mechanisms of attentional capture and suppression of a singleton distractor.

[Zehetleitner et al. \(2013\)](#) drew a similar conclusion, finding that the salience between a singleton target and singleton distractor could influence the distractor interference effect in a behavioral and eye-tracking study. They asked participants to search for a tilted target bar and ignore a bright distractor. The salience difference condition results from the change between the target's orientation and the distractor's luminance. Behavioral results showed that, regardless of whether the distractor was approximately salient than the target, RT interference existed and was caused by a competing distractor capturing attention. They concluded that each salience value follows a specific selection time distribution, and that attentional capture occurs when the selection time distributions of the target and distractor overlap. Hence, selection is stochastic, and attentional capture occurs with a probability that depends on relative saliency. However, RT interference is an indirect measure of the order of attentional selection, and the underlying mechanism can only be inferred. Our study provides direct electrophysiological measurements of the order and magnitude of N2pc, and P_D influenced by relative saliency. Moreover, we proposed an attentional processing model based on three levels of relative saliency between singleton distractors and singleton targets (i.e. low-, medium-, and high-level; see [Fig. 5B and C](#)).

Comparison with the three-stage model proposed by Chen et al. (2023)

[Chen et al. \(2023\)](#) proposed a three-stage model based on the physical salience of stimuli in a visual search when the task involved a singleton detection mode. According to this model, salient distractors undergo an early perceptual stage (Ppc), followed by an attentional capture stage (N2pc), and finally, a distractor suppression stage (P_D). The variable that regulates these stages is the physical salience of the distractors. The salience of the distractors likely influences the amplitude and latency of the corresponding components in each stage. This means that the more salient the distractors, the stronger the early perception,

attentional capture, and suppression, resulting in larger amplitudes of the corresponding ERP components. Conversely, lower saliency may lead to smaller amplitudes or even the absence of these stage-specific components.

Consistent with the three-stage model, the results of this study also revealed that attentional processing of a singleton distractor initially involves attentional capture, followed by attentional suppression. Furthermore, the relative saliency between a singleton distractor and a singleton target was found to influence the attentional processing of the singleton distractor. Specifically, higher relative saliency of a singleton distractor in the search array was associated with larger amplitudes of components in response to attentional capture and suppression. Conversely, lower relative saliency resulted in smaller amplitudes or even the absence of these components.

It should be noted that in [Chen et al. \(2023\)](#), the saliency of different distractors (i.e. onset vs. color) was indirectly compared with a shape target in a search array, whereas in the present study, the saliency of face and color was directly contrasted as a distractor or a target in a search array, competing for attentional resources. Several distinctions between the current study and [Chen et al. \(2023\)](#) should be noted.

First, this study clarifies the concept of relative saliency, proposing that relative saliency is the saliency contrast between a singleton target and a singleton distractor. We also manipulated the relative saliency level between singleton targets and distractors.

Second, the present study did not show the Ppc component, which is likely associated with imbalances in low-level sensory properties of the stimuli across hemispheres, representing early perceptual processing ([Gaspelin et al. 2023](#)). Prior studies on the attentional processing of distractors have also failed to observe this component using a color singleton ([Gaspelin and Luck 2018](#); [Stilwell and Gaspelin 2021](#)), which implies that the early perceptual stage may not be necessary for attentional processing, or its presence may depend on the nature of the stimulus. Some studies have shown that Ppc is sometimes reduced when the distractor is less salient ([Kerzel and Huynh 2022](#); [Drisdelle and Eimer 2023](#); [Gaspelin et al. 2023](#)). In our study, the saliency of the face and color distractors was perhaps not as high as the onset distractors observed in [Chen et al. \(2023\)](#), such that they were likely insufficient to elicit the contralateral Ppc component at the early stage (see also [Gaspelin and Luck 2018](#); [Stilwell and Gaspelin 2021](#)).

Third, the three-stage hypothesis assumes that saliency consistently and synchronously impacts attentional capture and suppression. However, in the present study, we found that when a face was used as a singleton distractor, due to the automatic processing of the face, relative saliency did not affect the attentional capture and suppression processes of a face singleton distractor in a monotonous manner. Specifically, relative saliency only affected the attentional suppression of a face distractor but did not impact its attentional capture. This specificity of face stimuli in attentional capture and suppression is addressed in the next section.

Specificity of faces in attentional capture and suppression

Faces may undergo more automatic processing than feature-based color stimuli ([Golan et al. 2014](#); [Prunty et al. 2023](#)). Attentional capture by a face singleton target resists the fluctuation of a color singleton distractor's saliency. There were no significant differences in N2pc amplitudes elicited by face singleton targets across the three experiments, indicating that attentional capture

by a face singleton target was irrelevant to the decrease in its relative saliency. These results are consistent with previous research, which showed that faces consistently “pop out” and exhibit a pop-out effect as target stimuli, regardless of the number or brightness of background distractors ([Golan et al. 2014](#); [Prunty et al. 2023](#)).

However, there is still controversy about which of the low-level perceptual types of information extracted from faces or the high-level social value of faces themselves could account for this face specificity. Several studies have emphasized the role of the low-level inner properties of faces in the rapid detection of faces. For example, [Pereira et al. \(2020\)](#) controlled extraneous factors that potentially bias attention, like luminance, and required participants to perform a dot-probe task with images of a face, house, and neutral cues. The results showed no evidence of attentional biasing effects on faces (see also [Guyader et al. 2017](#); [Pereira et al. 2022](#)). Although the current study did not compare the attentional processing of faces with that of color in a condition that controls all extraneous physical factors that potentially bias attention; however, some valuable insights about the flip side can be drawn from the current results. The faces used in the current study were displayed as gray-level versions with a strict control of luminance that matched that of the color stimulus. Thus, the possibility that the special attention processing of faces resulted from such color or luminous contrast can be ruled out. Furthermore, in Experiment 3, while blurred versions of faces with fewer physical details were used (despite being able to still be perceived as a face), there was no evidence that the amplitude of the N2pc component elicited by this type of face singleton target differed from that in Experiments 1 and 2.

Therefore, the result of Experiment 3 further excludes the decisive role of the low-level properties in the specificity of faces in attentional capture, which is consistent with previous research that has shown asymmetrical results when comparing the attentional processing of faces with the processing of other materials sharing equivalent low-level properties ([Langton et al. 2008](#); [Morand et al. 2010](#); [Negrini et al. 2017](#)).

When a face was presented as a singleton distractor, the current results showed that the attentional suppression process for faces did not elicit as large a P_D component as that observed for abrupt onset stimuli in [Chen et al. \(2023\)](#). This may be because the saliency of a face singleton distractor is not as high as that of an abrupt onset distractor, requiring fewer cognitive resources for suppression. Moreover, unlike a color singleton distractor, the N2pc component elicited by a face singleton distractor appears bimodal. Prior research has also found that N2pc elicited by a face singleton distractor can be divided into early and late stages ([Holmes et al. 2009](#); [Wójcik et al. 2019](#)). For example, [Holmes et al. \(2014\)](#) found that a higher working memory load resulted in greater attentional capture by threatening faces, eliciting an early N2pc (180 to 252 ms) representing attentional capture and a late N2pc (252 to 320 ms) representing attentional maintenance.

Limitations

The current study has several limitations. First, the current study cannot completely rule out the contribution of low-level physical properties to a face attention capture effect. When a face singleton was presented as a target, there was no difference between the three experiments. When a face singleton was presented as a distractor, there was no difference in the amplitude of the N2pc elicited by it between Experiments 1 and 2. However, the blurred face singleton distractor in Experiment 3 seemed to interrupt its

attentional processing specificity because it could no longer be rapidly recognized and did not capture attention. Task relevance may have played a role in altering facial salience. Conversely, except for brightness and color contrast, not all other extraneous physical factors that potentially bias attention was controlled for faces. Prior research has also indicated that the size relationship between stimuli can influence the degree of attentional capture (Luck and Hillyard 1994; Infanti et al. 2017). It is possible that this face specificity is partly derived from the differences in the manipulation of the color ring and face size in the experimental setup, which need to be further accounted for in future research. In sum, assertions about the special attentional capture effects of faces should be treated with caution, as comparing the salience of information across two dimensions is subject to various factors.

Second, the experimental design used a singleton detection paradigm in visual search, where the stimuli in each search array were predominantly homogeneous and the target was an odd item in one dimension (e.g. Lin et al. 2024). Such an experimental design encouraged participants to adopt a singleton detection mode, in which a singleton distractor is more likely to capture attention, making it potentially challenging to actively suppress a color distractor (Gaspar et al. 2016). Thus, future research should consider employing alternative search paradigms, such as feature search paradigms, to investigate the effects of relative saliency on attentional capture and suppression of the distractor.

Third, in this study, both faces and colors have relatively high physical salience, allowing them to capture attention as either a target or a distractor. It is crucial to note that when exploring the impact of relative saliency, it is not considered in isolation; task relevance and physical energy must also be considered in future research. For instance, in Fig. 5A, the gray 45° dotted line represents the isopleth line, where the singleton target and distractor have equal salience. All points on the isopleth line are mapped to the central black point on the relative saliency coordinate. Importantly, even if the relative saliency is the same, attentional processing toward different singleton distractors may not be the same. For instance, both the singleton distractor and target may have extremely low salience, as shown by the position of the white triangle in Fig. 5A, in which case the task-irrelevant singleton distractor may be directly ignored. Alternatively, both the singleton distractor and target may have extremely high salience, as shown by the position of the gray triangle in Fig. 5A. In this case, even if the singleton distractor is task-irrelevant, its high physical energy may capture attention or be directly suppressed. Future research should consider factors other than relative saliency, such as physical salience, social salience, and cognitive control (e.g. Wang et al. 2019), to explore their impact on the attentional process to a singleton distractor.

Finally, we only manipulated the degree of salience between a singleton distractor and a singleton target and observed its impact on attentional processes, without employing a more direct measure of salience between experiments. Future research could be conducted using more objective measurement methods, such as psychophysics or computational modeling (Stilwell et al. 2022; Stilwell et al. 2023), to further explore and validate the hypotheses of this study.

Conclusion

In conclusion, we demonstrate that relative saliency between a singleton distractor and a singleton target affects attentional processing to perceptual color and social face stimuli when they directly compete for attentional resources in a search array. The

relative saliency between a singleton distractor and a singleton target influences the attentional capture and suppression processes of the distractor. The higher the relative saliency of the singleton distractor, the stronger the attentional capture and suppression associated with that distractor, and vice versa. Finally, a face target/distractor shows a certain degree of specificity in attentional capture and suppression, possibly because of its high social salience and automation-related processing.

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Author contributions

Yue Zhang (Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing—original draft, Writing—review & editing), Hai Zhang (Data curation, Validation, Writing—original draft), and Shimin Fu (Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing—review & editing).

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